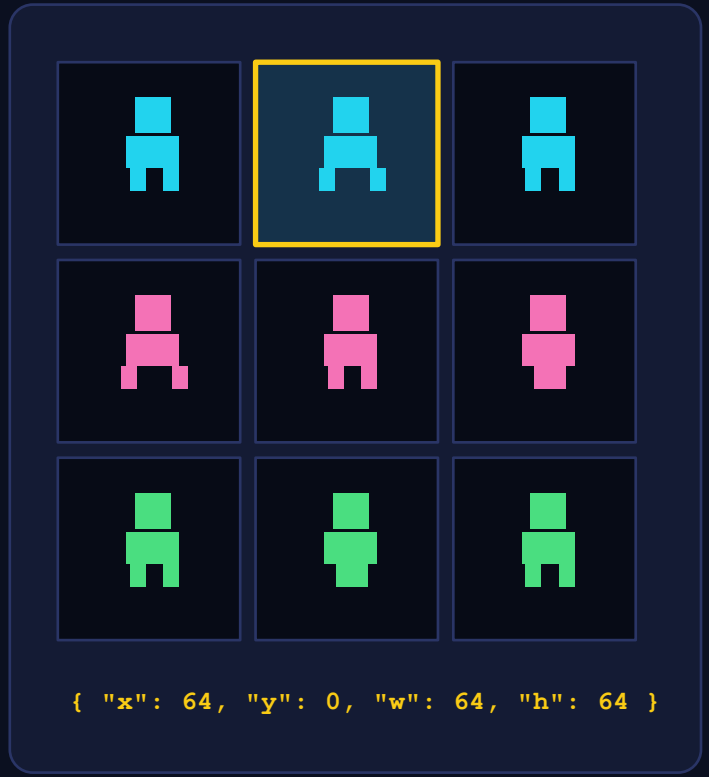
CCA GAME DEVELOPMENT

Sprite Sheets & JSON

Module 3 — Session 9

World Building & Camera Systems

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LEVEL 01 · TODAY'S GOAL

Today's Goal

By the end of this session, you will be able to:

1. Understand how sprite sheets organize animation frames.
2. Read sprite information from a JSON file.
3. Extract individual sprites using Pygame.
4. Create a simple sprite sheet parser.



TODAY'S MISSION

Turn **one large sprite sheet** into **multiple individual sprites**!

What Are Sprite Sheets?

A **Sprite Sheet** is one image containing multiple smaller images called sprites or animation frames.

Instead of saving every animation frame separately, we can organize them in one image.

WHY USE SPRITE SHEETS?

- Keep animation frames organized.
- Make sprite management easier.
- Reduce the number of separate image files.

CHARACTER SPRITE SHEET

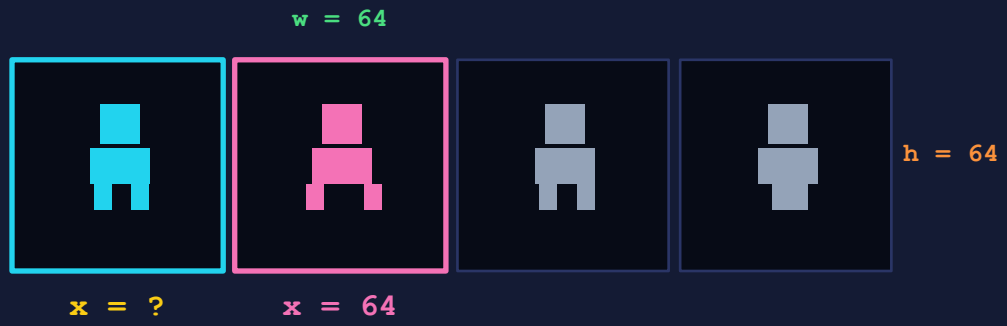


Example: walking, jumping, and attacking in one sheet.

Understanding Frame Coordinates

To extract an individual sprite, we need to know where it is located inside the sprite sheet. We use four important values:

- **x** — Horizontal position.
- **y** — Vertical position.
- **w** — Frame width.
- **h** — Frame height.



Example: a frame starting at `(0, 0)` with a size of `64 × 64` pixels.

Quick Question: If the second frame starts at x = 64, where does the first frame start?

LEVEL 04 · JSON

What Is JSON?

JSON (JavaScript Object Notation) is a format used to store structured information.

For sprite sheets, JSON stores information about each frame, such as its name, position, and dimensions.

```
{
    "frames": {
        "idle_1": {
            "x": 0,   "y": 0,
            "w": 64, "h": 64
        },
        "idle_2": {
            "x": 64, "y": 0,
            "w": 64, "h": 64
        }
    }
}
```



JSON = A MAP

Think of JSON as a map that tells Python where each sprite is located!

`idle_1 → (0, 0)`

`idle_2 → (64, 0)`

Reading JSON in Python

Python has a built-in `json` module that allows us to read JSON files.

First, import the module and load the file:

```
import json

with open("assets/sprites.json", "r") as file:
    data = json.load(file)

frames = data["frames"]
```

WHAT HAPPENS HERE?

`json.load()`

reads the JSON file.

`data`

stores the information.

`frames`

accesses the sprite frame data.

Loading & Extracting Sprites

First, load the original sprite sheet:

```
sprite_sheet = pygame.image.load(
    "assets/spritesheet.png"
).convert_alpha()
```

Then, extract an individual frame using its JSON information:

```
info = frames["idle_1"]

rect = pygame.Rect(
    info["x"], info["y"],
    info["w"], info["h"]
)

frame = sprite_sheet.subsurface(rect)
```

HOW DOES IT WORK?

1. Get the frame information from JSON.
2. Read its position and dimensions.
3. Create a rectangle around the selected area.
4. Extract the sprite using subsurface().

Result: one individual sprite extracted!

Automating Sprite Extraction

What if our sprite sheet contains 20 or more frames?

Instead of extracting them one by one, we can use a loop.

```
sprites = {}
for name, info in frames.items():
    rect = pygame.Rect(
        info["x"], info["y"],
        info["w"], info["h"]
    )
    sprites[name] = sprite_sheet.subsurface(rect)
```

WHAT'S HAPPENING?

`frames.items()` goes through every frame in JSON.

`name` stores the frame name.

`info` stores its coordinates and dimensions.

`sprites` stores all extracted frames in a dictionary.

Result: All frames are extracted automatically!

LEVEL 08 · LET'S CODE!

Mini Mission: Build a Sprite Sheet Parser

YOUR TASKS

1. Load the provided sprite sheet.
2. Read its matching JSON file.
3. Extract individual frames using `subsurface()`.
4. Use a loop to extract all frames automatically.
5. Display the extracted sprites in a Pygame window.

OPTIONAL CHALLENGE

- Display several extracted frames side by side.
- Try using a different sprite sheet.
- Organize extracted frames by animation name.



Remember: The JSON file and sprite sheet must match!

NEXT SESSION

Tilemaps & World Building

Let's learn how to build larger game worlds using tiles!

